CSC 610 – Independent Research

Title: Determinants of Global Life Expectancy: A Country – Year Panel Analysis

Student: Nakita Ramachandra Purohit

Mentor: Dr. Md Amir Amiruzzaman

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ABSTRACT

This study examines the socioeconomic, health system, and environmental determinants across the several countries over time. In this study Country-Year panel dataset was constructed which covers 233 countries from 2000-2023 which yields 5,592 observations by combining publicly available data from sites like World Bank world development indicators and Our world in data. The outcome variable here is life expectancy at birth in years. The main indicators includes GDP per capita that is log-transformed, government expenditure on education as a percentage of GDP, current health expenditure as a percentage of GDP, and PM2.5 mean annual air pollution exposure. Life expectancy separated by sex is just used for descriptive comparison. The analysis combines a set of visualizations with three sequential regression models of increasing complexity. A fixed effects specification was considered but is not implemented in this version; it is listed as future work. A secondary descriptive comparison of pre-2020 and 2020–2023 patterns is included to summarize the COVID-19 disruption and recovery. Results are interpreted as associations, not causal effects.

Keywords: life expectancy, panel data, determinants, regression, GDP per capita, health expenditure, education, PM2.5, COVID-19

1. INTRODUCTION

Life expectancy at birth is widely cited and is used as one of the indicators of population health. It shows how the mortality rates adds up across all age groups and it helps to track the progress in health and development across all countries over the time. Life expectancy has increased globally over the last century from around 32 years in 1900 to more than 70 years by early 2000s, by 2019 it was around 73 years (Riley, 2005; Dattani et al., 2023). There are still differences in life expectancy around the world. In high-income countries it exceeds 80 years but in many low-income countries it’s still below 65 years, this shows differences in economic resources, health systems, environmental conditions and behavioral risk profiles (World Bank. (n.d.); Cao et al., 2020).

The main framework for understanding variation in life expectancy was established by Preston (1975), who documented a strong non-linear positive relationship between the national income per capita and life expectancy for decades across the countries. Preston curve showed that increase in life expectancy from more income are largest at low-income level and reduces at higher ones. Also importantly, Preston found out that the curve started shifted upward over the time means countries at equal income level also achieved longer life expectancy in later time than the earlier ones. This showed that the factors beyond income including medical technology, sanitation infrastructure, nutrition and the spread of public health knowledge were independently contributing to life span gain worldwide.

Future research expanded this framework significantly. Scholars found that education, health system capacity, environmental quality and behavioral risk factors as the additional factors of life expectancy that income cannot fully explain (Kabir, 2008; Lutz & Kebede, 2018; Chetty et al., 2016). The COVID-19 pandemic disrupted the long term trend. Many countries observed sharp decline in 2020 and 2021, in some cases it was reversed a decade or more of progress within a single year (Heuveline, 2022; Schöley et al., 2022). This has caused a interest in understanding the factors that affect population health. Countries with stronger health systems, high income, and more educated population showed greater resilience during the pandemic period.

So, this study contributes to this literature by creating a reproducible, country-panel dataset of 233 countries from year 2000 to 2023 and it systematically measures the relationship focused set of factors and life expectancy at birth. The main question research is: which key determinants are most strongly associated with cross-country and over-time variation in life expectancy at birth when considered together? A secondary extension comparing pre-2020 and 2020-2023 patterns is included to characterize the COVID-19 disruption and partial recovery.

2. RELATED WORK

2.1 Income and Economic Development

After the analysis of Preston in 1975 relationship between national income and life expectancy has become important to population health literature. Preston has used cross-sectional data over decades and documented that wealthier nations has longer life span consistently but the relationship was concave meaning that it got weaker at high levels. He also showed that the curve was shifted upward over the time on own, attributing this to external health improvements rather than the income. Bosworth and Keys (2004) extended this analysis further in a global context. Documented that while income growth is still the important factor long life span in low-income countries, the strength if this association weakened at high income levels this was similar to Preston's original study. Recently, Agrawal (2024) examined the link between GDP and life expectancy using the recent cross - country data and found that positive link still holds true with current data, but it varies a lot between regions and income groups. Chetty et al. (2016) showed the evidence from the United States that income - longevity gradient works not only between nations but also within the countries. For example, richest Americans live much longer than the poor on average.

2.2 Health Expenditure and Health System Quality

A number of studies have examined on how investing in health systems can help people live longer. Jaba et al. (2014) has analyzed OECD countries using the fixed effects regression and found significant positive effects of per capita health expenditure on the life expectancy. Linden and Ray (2017) analyzed 34 OECD countries from year 1970 to 2012 using panel time series method and found out that public health expenditure has strong positive effects on life expectancy compared to private spending, especially among countries with higher level of public expenditure as share of GDP. Anwar et al. (2023) used system GMM estimation for 38 OECD countries from 1996 to 2020 and showed that government health expenditure has positive impact on life expectancy, while air pollution has negative impact. Galvani-Townsend et al. (2022) took broader approach and found that countries with higher levels of public health coverage attained better life expectancy results after adjusting for income and other confounding factors.

2.3 Education

Education is recognized as the independent factor of increasing life expectancy, working through several pathways including high income, better health literacy, healthy behavioral choices and good use of preventive health care. Kabir (2008) analyzed 91 developing countries and found that adult literacy was one of the strongest predictors of the life expectancy. Lutz and Kebede (2018) used panel of 174 countries from 1970 to 2010 and showed that having education had significant positive impact on life expectancy even after accounting for GDP per capita. Aytemiz et al. (2024) analyzed OECD countries from 2005 to 2021 using second-generation cointegration methods and found that education and health expenditure had significant long term positive effect on life expectancy along with GDP per capita playing major role. The authors confirmed that these three factors together shows a share of longevity differences across countries.

2.4 Air Pollution and Environmental Risk

Environment quality especially air pollution, has gained lot of attention as determinant of population health outcomes. The State of Global Air (2025) has estimated that exposure to PM2.5 is one of the highest risk factors for global disease burden and life expectancy loss especially in South Asia, East Asia and Sub-Saharan Africa. Rahman et al. (2022) analyzed around 31 heavily polluted countries from 2000 to 2017 and found that PM 2.5 had a significant negative effect on life expectancy, alongside positive link with GDP and health expenditure. Anwar et al. (2023) also found similar negative effects of air pollution on health outcome within an OECD panel, strengthening the importance of environmental factors in models of population health.

2.5 COVID-19 and Recent Disruptions

The COVID-19 pandemic showed the severe mortality shock in decades globally. Heuveline (2022) documented serious declines in life expectancy in 2020 and 2021 across most of the world, major losses was in Eastern Europe, Latin America and some parts of Asia. Schöley et al. (2022), showed that while some of high-income countries with stronger health systems recovered by 2021 but many middle-income countries experienced below-trend life expectancy through 2022. Cao et al. (2020) had earlier projected the global and regional life expectancy through 2025, but projects were gradually disrupted by the pandemic, underscoring the long-term trends to acute health shocks. These findings motivates to include post 2020 period as secondary descriptive extension in this study.

3. DATA

3.1 Unit of Analysis and Timeframe

This study uses a country-year panel dataset for analysis, where each observation shows one country in one year. The dataset covers 233 countries over 24 years from year 2000 to 2023, producing 5,592 country-year observations before accounting the missing values. Here year 2000 was selected as the starting point because it aligns with the adoption of Millennium development goals and marks this period as substantially improved data availability for low and middle income countries. It should be noted that PM2.5 data is available through 2020 and GDP per capita through 2022; the analysis for these variables reflects their respective coverage ranges, as documented in the Table 2.

3.2 Data Sources

Data is collected from two important publicly available sources. The World Bank World Development Indicators (WDI) provided the four variables life expectancy at birth, government expenditure on education as the percentage of GDP, health expenditure as a percentage of GDP and PM2.5 air pollution exposure mean value. These files were directly downloaded from the World Bank Open Data portal (https://data.worldbank.org). Specific indicator codes and direct URLs are reported in Table 1. GDP per capita and life expectancy separated by sex are obtained from Our World in Data (https://ourworldindata.org), which uses World Bank and UN sources. All datasets are freely available and publicly accessible.

3.3 Variables

The outcome variable is life expectancy at birth in years (World Bank indicator SP.DYN.LE00.IN) which is defined as the average number of years a newborn is expected to live if current age-specific mortality rates remain constant throughout life. The primary factor to determine this are GDP per capita (PPP, international-$) from the OWID grapher dataset which is log -transformed using natural logarithm which is a standard transformation to compress wide income range and linearize the Preston Curve relationship. Government expenditure on education as percentage of GDP (World Bank indicator SE.XPD.TOTL.GD.ZS), serving as a substitute for public investment in the human capital, Health expenditure as a percentage of GDP (World Bank indicator SH.XPD.CHEX.GD.ZS), Capturing the national resources allocated to health system and PM2.5 mean annual exposure in microgram per cubic meter(World Bank indicator EN.ATM.PM25.MC.M3) showing air pollution as the environmental risk factor. Life expectancy divided by sex is just included as a additional variable for visualization only

3.4 Data Preparation and Cleaning

All six raw files were loaded and processed in Python using the pandas library. World Bank files were delivered in wide format with one row per country and one column per year; these were reshaped into long format using the melt function, producing one row per country-year observation. Our World in Data files used the column name "Code" for the country identifier, which was renamed to "Country Code" to match World Bank files before merging. All six datasets were merged sequentially using a left join on the combination of Country Code and Year, with life expectancy as the backbone. Rows corresponding to regional or income-group aggregates were identified by their non-standard ISO codes and removed, retaining only individual country observations. The final dataset covers 233 countries and 24 years. GDP per capita was log-transformed using the natural logarithm. Missing values were retained as NaN and are reported by variable in Table 2. Sample sizes are reported separately for each regression model to reflect variable-specific coverage.

Table 1. Variable Dictionary

Variable	Definition	Unit	Source	Indicator Code	URL
Life expectancy	LE at birth, total	Years	World Bank	SP.DYN.LE00.IN	https://data.worldbank.org/indicator/SP.DYN.LE00.IN
GDP per capita	GDP per capita, PPP	Intl. $	OWID/ World Bank	-	https://ourworldindata.org/grapher/life-expectancy-vs-gdp-per-capita
Log GDP	Natural log of GDP per capita	Log Intl. $	Derived	-	-
Education expenditure	Govt. expenditure on education	% of GDP	World Bank	SE.XPD.TOTL.GD.ZS	https://data.worldbank.org/indicator/SE.XPD.TOTL.GD.ZS
Health expenditure	Current health expenditure	% of GDP	World Bank	SH.XPD.CHEX.GD.ZS	https://data.worldbank.org/indicator/SH.XPD.CHEX.GD.ZS
PM2.5	Mean annual PM2.5 exposure	$\mu g/m^3$	World Bank	EN.ATM.PM25.MC.M3	https://data.worldbank.org/indicator/EN.ATM.PM25.MC.M3
LE women	LE at birth, female	Years	OWID	-	https://ourworldindata.org/grapher/life-expectancy-of-women-vs-life-expectancy-of-men

LE men	LE at birth, male	Years	OWID	-	https://ourworldindata.org/grapher/life-expectancy-of-women-vs-life-expectancy-of-men

Table 2. Missing Data Summary (generated directly from master_panel.csv)

Variable	Years Available	Missing (%)	Countries (any data)	Countries (80%+ coverage)	Countries (complete)
life_expectancy	2000-2023	0.43	232	232	232
edu_expenditure_pct_gdp	2000-2023	35.12	215	110	20
health_exp_pct_gdp	2000-2023	12.37	208	202	198
pm25	2001-2020	23.1	215	215	215
gdp_per_capita	2000-2022	32.55	164	164	164
le_women	2000-2023	7.73	215	215	215
le_men	2000-2023	7.73	215	215	215

Table 2: *Missing (%) calculated across full 2000 - 2023 panel. Coverage thresholds computed against each variable's own available year range. PM2.5 available 2001 - 2020; GDP per capita available 2000–2022*

4. METHODS

4.1 Visual Analysis

The analysis starts with set of descriptive visualizations before any statistical modeling to explore global and regional patterns. The first figure shows the average global life expectancy from 2000 to 2023 including the COVID-19 dip in 2020-2021. The second figure shows the disaggregated trend by geographic region showing how different parts of the world have converged or diverged over the period of time. The third one is a scatter plot of life expectancy against the log GDP per capita which is Preston curve , pooled across the country-years with a fitted line to characterize the shape of the income and health relationship in the data. The fourth one is a correlational heatmap showing the pairwise links among all the major key variables, which informs multicollinearity assessment in the regression stage. The fifth compares life expectancy trends for men and women over the time. The sixth figure compares life expectancy trends before and after 2020 to visualize the COVID-19 impact and subsequent recovery.

4.2 Regression Models

Three sequential OLS regression models were estimated to quantify the relation between the selected determinants and life expectancy at birth. All models were estimated on the combined country-year panel. Model 1 is the baseline specification and regress life expectancy on log GDP per capita and a linear year trend, replicating the Preston Curve in a regression framework. Model 2 extends Model 1 by adding government expenditure on education and current health expenditure as a percentage of GDP, testing whether public investment in human capital and health systems is associated with life expectancy beyond what income alone explains. Model 3 further adds PM2.5 mean annual exposure as an environmental risk variable. A fixed effects specification was considered, but it is not implemented in this version; it is listed as future work. Fixed effects models control for all unobserved time-invariant country-level characteristics as well as common year-specific shocks. All results are reported with coefficients, standard errors, and sample sizes, and are interpreted as associations rather than causal effects.

4.3 Robustness Checks

A number of robustness tests were undertaken to determine the stability of the key results. Firstly, all regressions were recalculated for the pre-2020 sub-sample in order to test the potential impact of the COVID-19 era on the coefficients. Secondly, a possible outlier sensitivity of GDP per capita was tested by examining regressions using both full and trimmed data. Thirdly, variance inflation factors were calculated for all independent variables to formally test for multicollinearity. Lastly, this additional re - estimation was not required because VIF values for Model 3 predictors were all below 2, indicating low multicollinearity

5. RESULTS

5.1 Visual Analysis

Six figures were produced to explore global and regional patterns in life expectancy before any statistical modeling.

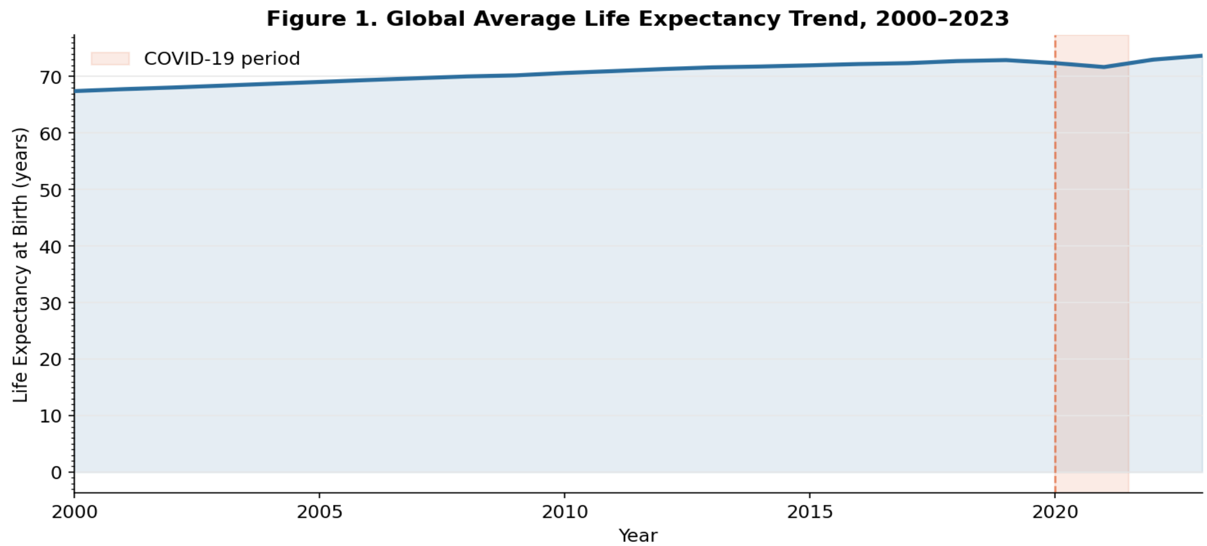

Figure 1 shows global average life expectancy from 2000 to 2023. Life expectancy has increased from 67.4 years in 2000 to 72.9 years in 2019 which is an increase of approximately 5.5 years over almost two decades. Life expectancy declined when COVID-19 pandemic hit globally, it declined from 72.9 years in 2019 to 72.4 years in 2020 and further dropped to 71.7 years in 2021. It started recovering in 2022 when average became 73.0 years and 73.7 years in 2023 which is slightly above life expectancy before pandemic.

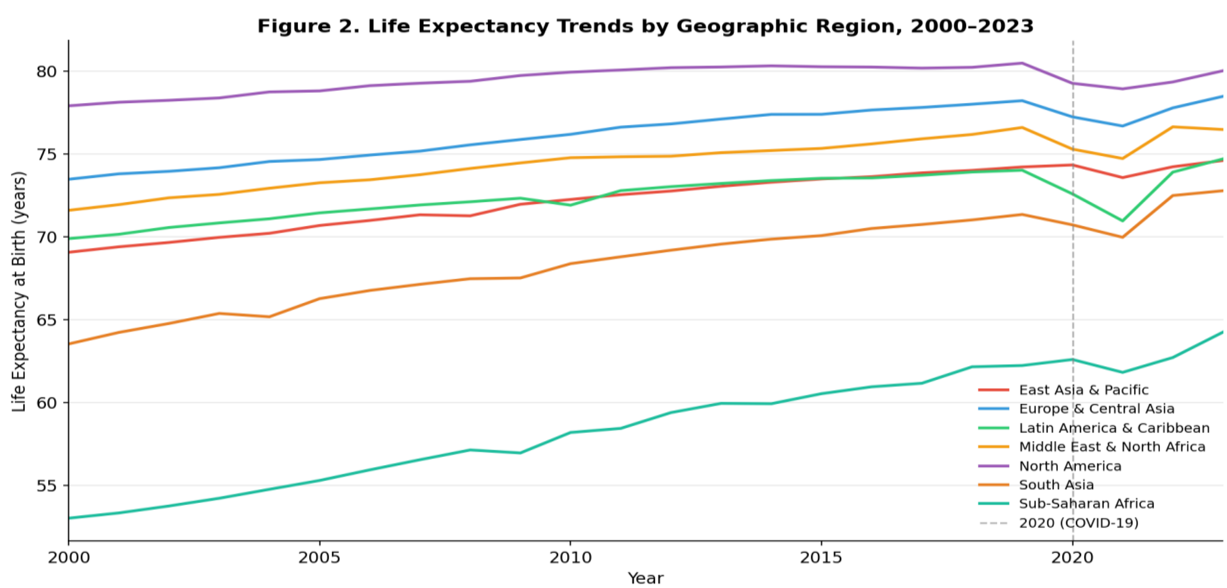

Figure 2 shows life expectancy trends for some selected geographic regions from the year 2000 to 2023. North America had the highest average life expectancy 79.3 years in 2022, next was Europe and Central Asia at 77.8 years and then the Middle East and North Africa at 76.6 years. Latin America and Caribbean and East Asia and pacific were at 74.0 and 74.2 years which is a middle range. Sub-Saharan Africa had the lowest average 62.7 years and it showed notable improvement over the study period. The COVID-19 dip is visible across most of the regions. Countries that could not be assigned to any named region are grouped as "Other"(69 countries in 2022) and they are not shown as separate line in this figure here.

Figure 3. Life Expectancy vs. Log GDP per Capita (Preston Curve), 2000–2022

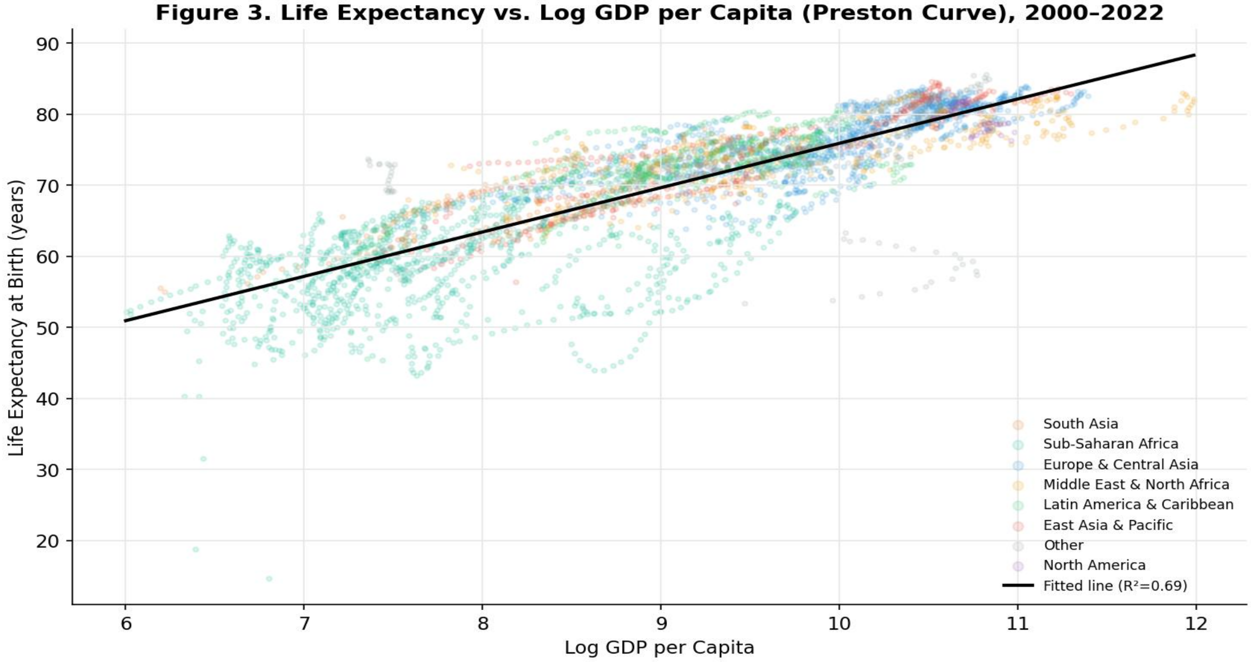

Figure 3 shows the Preston Curve plotting life expectancy versus log GDP per capita across all the available country – years. The scatter plot shows a strong positive relationship between income and longevity. The fitted line shows an R^2 of 0.694 means approximately 69% of variation in the life expectancy across country – years is explained by log GDP per capita alone and the slope of 6.24 indicates that one unit increase in the log GDP per capita is associated with 6.2 additional years of life expectancy on average approximately. So, considerable variable around the fitted line shows that other factors beyond income also plays and important role motivating the multivariate models estimated in Section 5.2.

Figure 4. Correlation Heatmap of Key Variables

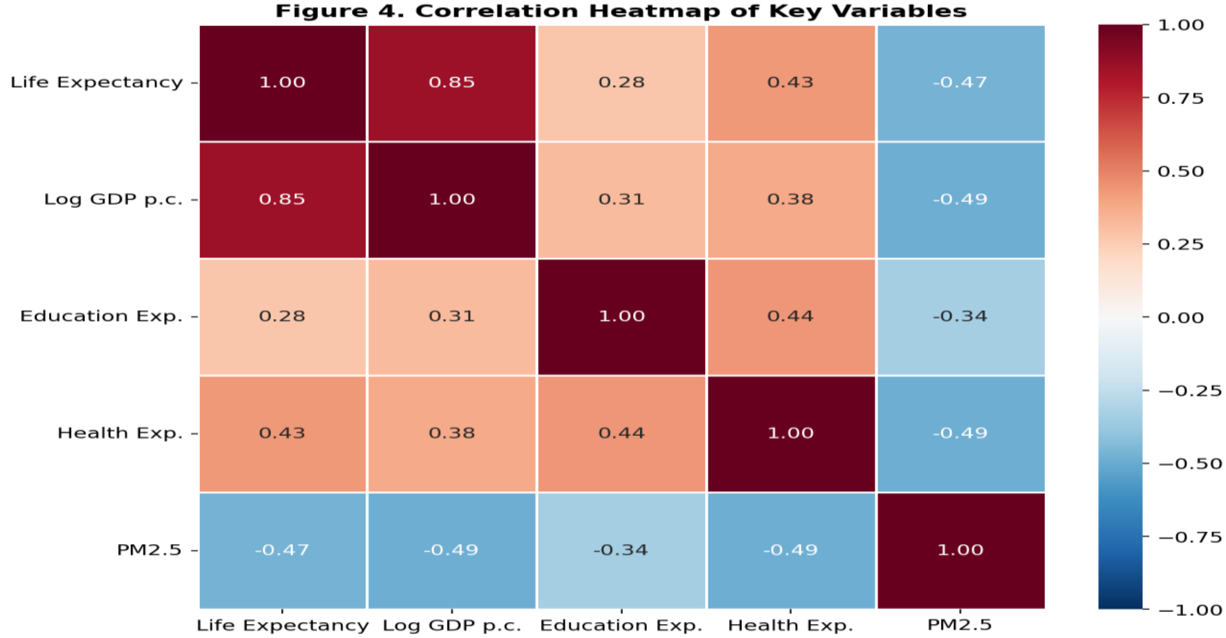

Figure 4 above shows the pairwise correlation matrix between the five important variables, based on 2,318 complete cases where all the five variables occur at once. Log GDP per Capita exhibits a very high positive correlation of 0.854 with Life Expectancy, supporting the Preston Curve phenomenon in the dataset. Health Expenditure and Education Expenditure shows positive correlation coefficients of 0.432 and 0.278 respectively. On the other hand, PM2.5 Air Pollution exhibits a strong negative correlation of -0.471 with Life Expectancy, means countries with high pollution level are expected to have shorter lifespans. The moderate correlations among predictors suggest multicollinearity is unlikely to be severe, though variance inflation factors were checked formally in the regression stage.

Figure 5. Life Expectancy Trends by Sex, 2000–2023

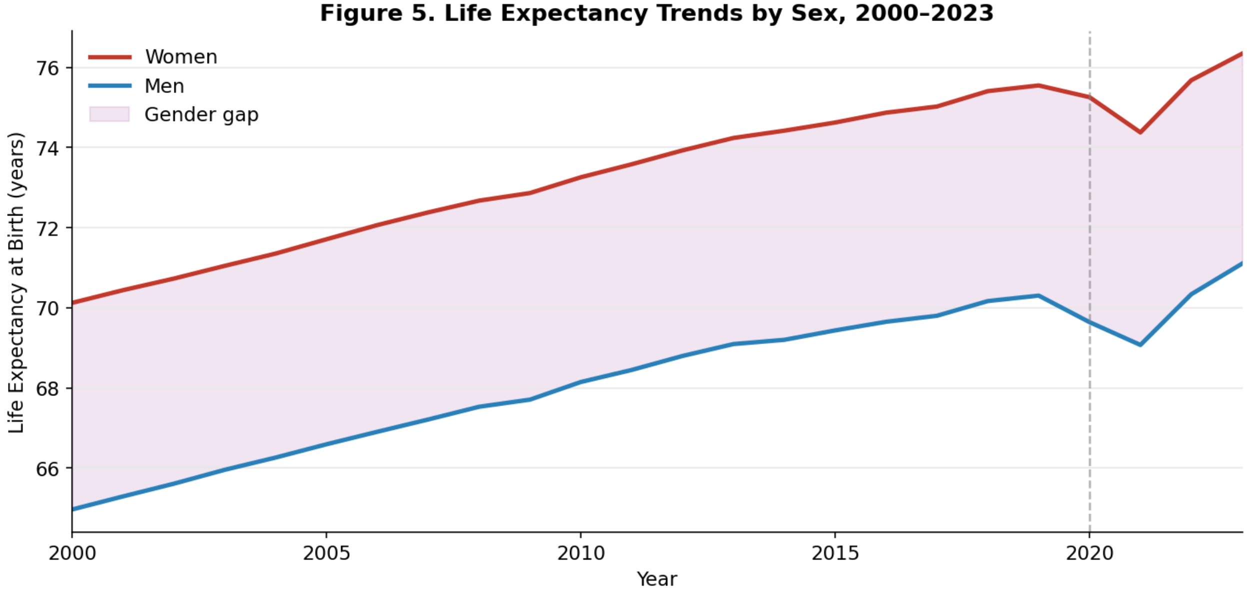

Figure 5 compares life expectancy trends for women and men from 2000 to 2023. Average life expectancy of women in 2000 was 70.1 years compared to 65.0 years for men, there was gender gap of 5.2 years. By 2023, women average life expectancy was 76.3 years and men 71.1 years with the same gap of 5.2 years, and it was stable throughout study period. Both men and women life expectancy showed a dip during 2020 and 2021 and started recovering by 2022 and 2023.

Figure 6. Distribution of Life Expectancy: Pre-2020 vs. 2020-2023

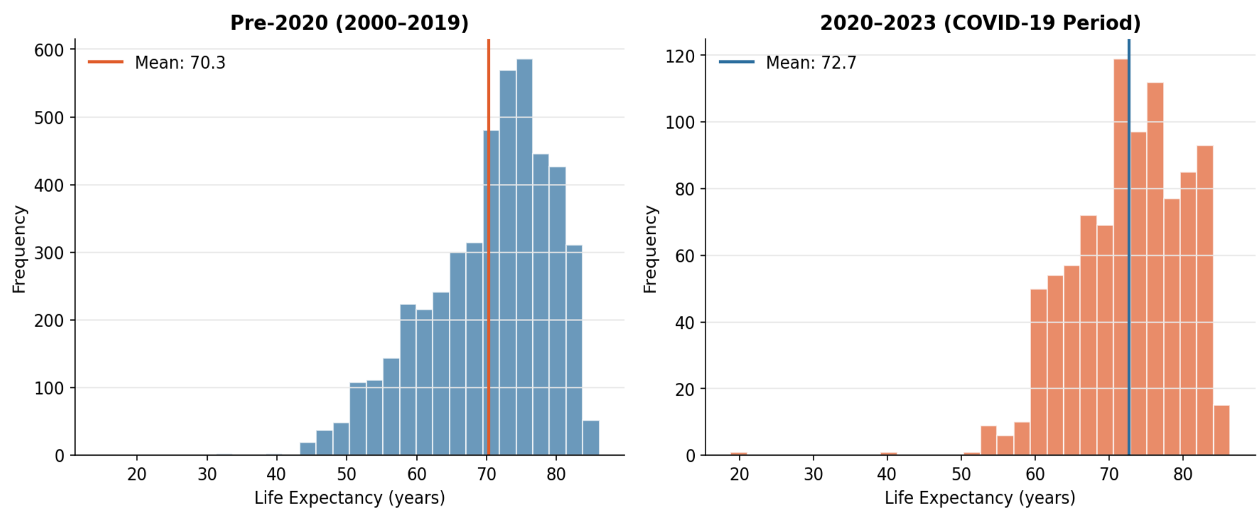

Figure 6 compares the distributions of life expectancies before and after 2020. The data available before 2020 which covered 2000 to 2019 had a mean life expectancy of 70.4 years across all the country – year observations and year 2020 to 2023 had mean life expectancy of 72.7 years. After 2020 the distribution has shifted toward right, this reflects long run upward trend despite COVID-19 disruption in the individual year values for 2020 and 2021. This difference shows the continuing long run upward trend and sample composition across country-year observations and it does not imply COVID increased life expectancy

5.2. Regression results

The findings from the three subsequent OLS regression analyses performed on the country-year panel data are presented in Table 3. Robust standard errors (HC1) were used in all regressions to control for heteroskedasticity.

In Model 1, life expectancy is regressed on log GDP per capita and the linear year trend, using 3,772 observations from 164 countries. Log GDP per capita has a significantly high positive coefficient of 6.15, implying that a one-unit increase in log GDP per capita leads to an average gain of 6.1 years in life expectancy while controlling for year trend. The coefficient of the year term, which is 0.13, depicts the overall improvement in life expectancy on a global scale irrespective of income changes. The two variables collectively explain 70.3% of the variation in life expectancy (R^2 = 0.703) which is consistent with the Preston curve relationship shown in the visualizations.

Model 2 further enhances baseline by introducing expenditure on education by the government and current health expenditure as a percentage of GDP. Data gaps in these variables reduce our sample size to 2,683 observations covering 155 countries. Like before, log GDP per capita proves positive and significant at 6.08, suggesting that the income - longevity connection still holds when other explanatory variables are included. Current health expenditure as a percentage of GDP is positive and significant, with a coefficient of 0.43, meaning that a one-point rise in health expenditure to GDP ratio results in about 0.43 extra life expectancy years, after controlling for income levels and the year trend. This result is supported by Jaba et al. (2014) and Anwar et al. (2023), who reported similar positive coefficients on health expenditure for life expectancy in OECD panel data. Government expenditure on education displays a negative and insignificant coefficient value of -0.14 (p = 0.115). However, this does not undermine the role of education in health. Instead, it reveals that government expenditure on education as a percentage of GDP is a poor indicator of education itself, as highlighted by other studies, including Lutz & Kebede (2018) . The R^2 increases to 0.747.

Model 3 includes PM2.5 (average annual exposure) and is estimated on the subsample of 2001 - 2020 because PM2.5 data is only available up to 2020 giving 2,318 observations for 155 countries. It must be mentioned that the reduction in the number of countries from 164 in Model 1 to 155 in Models 2 and 3 results from the increased data availability requirements resulting from adding new variables; the coefficients could have been affected by this sample modification rather than the inclusion of new independent variables. Log GDP per capita continues to prove its significance with a value of 6.00. Health expenditure is still significant with a value of 0.43. The variable PM2.5 presents a negative and statistically significant coefficient of -0.015 (p = 0.011). As it is seen, higher levels of PM2.5 air pollution are correlated with reduced life expectancy when accounting for such factors as country income, healthcare costs, and educational attainment. Although the coefficient seems low, 10 $\mu g/m^3$ higher PM2.5 concentration is associated with a reduction of almost 0.15 years in life expectancy - a considerable amount that is especially important for populations living in high levels of pollution. This finding is consistent with Rahman et al. (2022) and the State of Global Air (2025), both of which document significant negative effects of PM2.5 on population - level longevity. Education expenditure remains borderline insignificant at p = 0.057. The R^2 is 0.749, a modest increase over Model 2.

Table 3. OLS Regression Results: Determinants of Life Expectancy at Birth

Variable	Model 1 (Baseline)	Model 2 (+Edu +Health)	Model 3 (+PM2.5, 2001 - 2020)
log_gdp	6.148*** (0.067)	6.084*** (0.092)	6.001*** (0.098)
Year	0.131*** (0.012)	0.095*** (0.014)	0.112*** (0.016)
edu_expenditure_pct_gdp	-	-0.138 (0.088)	-0.180* (0.095)
health_exp_pct_gdp	-	0.428*** (0.047)	0.433*** (0.050)
pm25	-	-	-0.015** (0.006)
Intercept	-248.338*** (24.581)	-178.342*** (27.177)	-210.376*** (32.730)
R^2	0.703	0.747	0.749
Adj. R^2	0.703	0.747	0.748
Observations (N)	3,772	2,683	2,318
Countries	164	155	155

Note: Robust standard errors (HC1) in parentheses. Model 3 estimated on 2001–2020 subsample due to PM2.5 data availability. * p<0.1, ** p<0.05, *** p<0.01.

5.3 Robustness Checks

Three robustness tests were performed to confirm the stability of the major results.

First, models 1 and 2 were re-run on the pre-2020 subsample to see whether the presence of the COVID-19 years affects the outcomes. First, in model 1 with a pre-2020 sample (N=3,280; 164 countries), the impact of log GDP per capita remained very high (6.25), whereas R^2 changed only insignificantly (0.704 versus 0.703). Second, in model 2 with a pre-2020 sample (N=2,280; 154 countries), the coefficient for health expenditure remained significant at 0.47 , as well as for log GDP at 6.19. Meanwhile, education spending became marginally significant at p=0.049 in the pre-2020 years subsample, meaning that accounting for the current pandemic slightly decreases its positive effect on the outcome. Overall, the main findings are stable across the full and pre-2020 samples.

Second, the impact of outliers on the GDP data was investigated by removing the top and bottom 1% of GDP per capita figures and rerunning Model 1. The results from the sample excluding outliers (N = 3,696, 164 countries) had a log GDP value of 6.20 and an R^2 of 0.696, very close to those of the original sample, suggesting that the presence of outliers does not influence the results.

Third, multicollinearity among predictors was assessed using variance inflation factors for all variables in Model 3. All VIF values were below 10 and the maximum VIF score of 1.55 was reported for health expenditure; therefore, multicollinearity is unlikely to be an issue among all models.

6. DISCUSSION

Consistently across the models estimated in this study, GDP per capita emerges as the strongest individual indicator of life expectancy. In all three models, the variable log GDP per capita exhibits a coefficient of around 6.0 to 6.1, indicating that a country twice as wealthy as another would expect to see an extra four to five years added to its citizens' average life expectancy. This result is the clear replication of the Preston curve shown in Figure 3, and it accords with the pioneering work of Preston (1975), followed by later confirmations provided by Agrawal (2024) and Bosworth and Keys (2004). Moreover, the year trend variable, showing persistence as a statistically significant factor, indicates that gains in global longevity are also due to other trends like medicine, public health, and living standards over time besides income increases.

Health expenditure as a percentage of GDP stands out as the second determinant factor that shows a positive and significant relation in Models 2 and 3. One unit increase in health expenditure per GDP results in 0.43 years gain in life expectancy, even after considering income. This study finding is further supported by arguments of Galvani-Townsend et al. (2022) and Anwar et al. (2023), who assert that greater investment in health care systems results in higher public health than economic development alone can explain. In addition, this finding has implications for policy making, since it means that directing national resources to the health sector will result in increased life expectancy, particularly in lower-income countries where health systems remain underfunded.

However, the result regarding education expenditure is more complex. Government spending on education as a percentage of GDP is insignificant in full-sample models, although it is close to significance in the robustness check before 2020. However, this finding does not run counter to the general body of research linking education to health because it centers more on educational achievement than expenditures by governments on education. Lutz and Kebede (2018) and Kabir (2008) have shown that it is the extent of education people have actually received, rather than expenditure on education, that most strongly predicts longer lifespan. The spending measure used here is therefore a weak proxy and future work incorporating years of schooling data would better capture this pathway.

PM2.5 air pollution shows very small but significant negative relation with life expectancy in Model 3. An increase in 10 μg/m^3 of annual exposure to PM2.5 results in an estimated decline in life expectancy of about 0.15 years at population level. Although the effect is not pronounced in size, its practical importance cannot be ignored because many countries in South Asia and Sub-Saharan Africa have been found to be far from compliance with the guidelines provided by WHO on the average. Reducing air pollution in heavily exposed populations could therefore contribute meaningfully to closing the life expectancy gap between regions. The results of the research support those of Rahman et al. (2022), State of Global Air (2025).

The analysis of the impact of the COVID-19 period shown in Figure 6, together with the robustness tests, proves that the pandemic was truly disruptive for the trend of global longevity. The average decreased from 72.9 years in 2019 to 71.7 years in 2021 in line with Heuveline (2022) and Schöley et al. (2022). It is good to note that the improvement in 2023 up to 73.7 years is also promising, although it shows the presence of significant differences between regions. In addition, the stability of regression coefficients across the full sample and pre-2020 subsample shows that COVID-19 did not alter structural relationships between determinants and life expectancy and proves that the nature of relationships remained unchanged.

7. LIMITATIONS

A number of limitations associated with the study must be pointed out. For one thing, the availability of information differs significantly from variable to variable as well as among countries. Data on education spending, for example, are unavailable in approximately 35% of country - year observations, while those on GDP per capita are lacking in approximately 32%. Hence, regression analysis is conducted based on a selection of countries for which there is information available. If such a selection is non-random, that is, if only the poorest and least informative countries were excluded from consideration, then the estimated associations may not generalize to the full global sample.

Second, the study reports associations and does not establish causal effects. The OLS models estimated here are subject to omitted variable bias. Unobserved factors such as governance quality, cultural health practices, geographic disease burden, historical health infrastructure, and institutional quality may simultaneously influence both the predictors and life expectancy. Results should therefore be interpreted as descriptive associations rather than causal estimates.

Third, the education variable considered in the present research, government expenditure on education as a percentage of GDP is a weak measure of education attainment. It is important to emphasize that all previous studies have clearly demonstrated that only years of schooling, and not education expenditure, can be regarded as the channel through which education improves health. Future work should include attainment measures such as mean years of schooling from UNESCO or Barro-Lee data.

Fourth, PM2.5 data is only available through 2020, limiting the environmental analysis to the 2001 to 2020 subsample. The period after 2020 may show different patterns given pandemic-related reductions in industrial activity and subsequent recovery, and these cannot be captured with the current data.

Fifth, a country and year fixed effects specification was planned but not implemented in this version of the analysis and remains a direction for future work. Fixed effects models would better isolate within-country changes over time from cross-sectional differences between countries and would reduce concerns about omitted variable bias.

8. CONCLUSION AND FUTURE WORK

This study constructed a reproducible country-year panel dataset covering 233 countries from 2000 to 2023 by merging publicly available data from the World Bank and Our World in Data. The analysis combined descriptive visualizations with three sequential OLS regression models to examine which key determinants are most strongly associated with life expectancy at birth globally.

It is evident from the results that log GDP per capita is the strongest variable predictor of life expectancy, accounting for about 70% of cross-sectional variation without considering other variables, and validating the Preston Curve relationship for present panel data. Percentage of health expenditure to GDP is found to be a statistically significant independent predictor beyond income, where an increase in 1 percentage point would yield 0.43 extra years of life expectancy. PM2.5 air pollution is a significant predictor of life expectancy independent of income and health expenditure, stressing the importance of environmental conditions in improving life expectancy. Education expenditure, on the other hand, is not statistically significant in any model, which may be due to problems inherent with the proxy indicator.

The COVID-19 pandemic caused a visible disruption in global life expectancy trends, with the global average falling from 72.9 years in 2019 to 71.7 years in 2021 before recovering to

73.7 years by 2023. The structural associations between determinants and life expectancy remain stable before and after 2020, suggesting that the pandemic did not fundamentally alter the underlying drivers of longevity globally.

Several directions in which the results can be enhanced in the future include first using attainment variables like mean years of schooling as an alternative to expenditure on education since that will give an idea of the relationship between education and health better. Secondly, including fixed effects by country and year will allow for strong identification of changes that happen over time in each country. Lastly, expanding the years covered by the PM2.5 variable to beyond 2020 when new data is available will improve the environmental analysis done. Finally, including other factors, such as urbanization levels, water access, and obesity rates, may be beneficial for the model.

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APPENDIX

Appendix A: Code Index

The following scripts were used to conduct the analysis. All scripts must be run in the order listed below. Full code for each script is provided in Appendices B through F.

Script / File	Purpose	Outputs
data_cleaning_merge.py	Loads World Bank + OWID files, reshapes wide→long, merges to master panel	master_panel.csv
verify_stats.py	Prints verified statistics used in Figures 1- 6 write-up	Console stats
visualizations.py	Generates Figures 1 - 6	fig1.png to fig6.png
regression_models.py	Runs Models 1 - 3 with robust standard errors	regression_summary.csv
robustness_checks.py	Runs pre - 2020, trimmed GDP checks and VIF	robustness_results.csv, vif_model3.csv

Appendix B: Data Cleaning and Merging Code

The following Python script was used to clean, reshape, and merge all six raw data files into the master panel dataset. The full script is also submitted as a separate file

```
# Research: Determinants of Global Life Expectancy

# Script:   Data Cleaning & Merging


import pandas as pd
import numpy as np

# STEP 1: Helper function this step cleans the World Bank "wide" format to "long" format

def load_wb(path, value_name):
  """
  Reads a World Bank CSV, skips the 4 junk header rows,
  keeps only Country Code + year columns,
  reshapes from wide to long format.
  """
  df = pd.read_csv(path, skiprows=4, encoding='utf-8-sig')
```

```
   # Keeping only the country identifier and year columns (year columns are numeric
strings)
   year_cols = [c for c in df.columns if c.isdigit()]
   df = df[['Country Name', 'Country Code'] + year_cols]

   # melting here organizing messy columns into rows to have same format for all data
   df = df.melt(
      id_vars=['Country Name', 'Country Code'],
      var_name='Year',
      value_name=value_name
   )

   df['Year'] = df['Year'].astype(int)
   df[value_name] = pd.to_numeric(df[value_name], errors='coerce')

   return df[['Country Code', 'Year', value_name]]

# STEP 2: Loading & Cleaning each file

#Life Expectancy at Birth (World Bank) outcome is life expectancy at birth, total (years)
le_wb = load_wb('LE_at_birth.csv', 'life_expectancy')

# Education Expenditure (World Bank) outcome is government expenditure on education as
% of GD
edu = load_wb('education.csv', 'edu_expenditure_pct_gdp')

#Health Expenditure % GDP (World Bank) outcome is Current health expenditure as % of
GDP(indicator: SH.XPD.CHEX.GD.ZS)
health = load_wb('Health_new.csv', 'health_exp_pct_gdp')

# PM2.5 Air Pollution has different format
pm_raw = pd.read_csv('pm2_5.csv', encoding='latin-1')

# This file has TWO indicators stacked keeping only mean annual exposure
pm_raw = pm_raw[pm_raw['Series Name'].str.contains('mean annual', na=False)]

# Year columns in the file look like "2001 [YR2001]" so extracting just the 4-digit year
year_cols = [c for c in pm_raw.columns if 'YR' in c]
pm = pm_raw[['Country Code'] + year_cols].copy()
pm.columns = ['Country Code'] + [c.split(' ')[0] for c in year_cols]  # e.g. "2001"

# again reshaping wide to long
pm = pm.melt(id_vars=['Country Code'], var_name='Year', value_name='pm25')
```

```
pm['Year'] = pm['Year'].astype(int)
pm['pm25'] = pd.to_numeric(pm['pm25'], errors='coerce')  # ".." → NaN

# GDP per Capita (Our World in Data)
gdp_raw = pd.read_csv('life-expectancy-vs-gdp-per-capita.csv')

# OWID uses "Code" not "Country Code" like other files so renaming to match World Bank
files
gdp = gdp_raw[['Code', 'Year', 'GDP per capita']].rename(columns={
  'Code': 'Country Code',
  'GDP per capita': 'gdp_per_capita'
})

#Life Expectancy by Sex (Our World in Data)
sex_raw = pd.read_csv('life-expectancy-of-women-vs-life-expectancy-of-men.csv')

# Again same renaming "Code" to "Country Code" to match other files
sex = sex_raw[['Code', 'Year', 'Life expectancy of women', 'Life expectancy of
men']].rename(columns={
  'Code': 'Country Code',
  'Life expectancy of women': 'le_women',
  'Life expectancy of men': 'le_men'
})

gdp = gdp.dropna(subset=['Country Code'])
sex = sex.dropna(subset=['Country Code'])

# STEP 3: Merging all files into one master Dataset
# Staring Life expectancy as base or backbone here
# then LEFT JOIN each other variable on Country Code and Year.
# LEFT JOIN keeps all LE rows even if other variables are missing for that row.

df = le_wb.copy()

for other_df in [edu, health, pm, gdp, sex]:
  df = df.merge(other_df, on=['Country Code', 'Year'], how='left')

# STEP 4: Filtering to do analysis on samples

# Keeping only 2000–2023
df = df[(df['Year'] >= 2000) & (df['Year'] <= 2023)]

# Remove World Bank aggregate regions (not real countries)
# These codes represent regions/income groups, not individual countries
```

```
aggregate_codes = {
    'WLD', 'HIC', 'LIC', 'MIC', 'LMC', 'UMC',
    'EAP', 'ECA', 'LAC', 'MNA', 'NAC', 'SAS', 'SSA',
    'AFE', 'AFW', 'EAR', 'TEA', 'TEC', 'TLA', 'TMN',
    'TSA', 'TSS', 'OSS', 'PSS', 'PST', 'CSS',
    'IBD', 'IBT', 'IDA', 'IDB', 'IDX', 'OED', 'EMU'
}
df = df[df['Country Code'].str.len() == 3]        # ISO codes are only 3 letters
df = df[~df['Country Code'].isin(aggregate_codes)]  # dropping regional aggregates

# STEP 5: Created derived Variables

# Logging transform GDP per capita because GDP ranges from ~$600 to ~$150,000 (very skewed).
# So log scale compresses this range and linearizes the GDP–LE relationship,
# which is standard in economics/epidemiology literature (Preston curve).
df['log_gdp'] = np.log(df['gdp_per_capita'].replace(0, np.nan))

# STEP 6: Last step is reporting and saving
print("=" * 60)
print("MASTER DATASET SUMMARY")
print("=" * 60)
print(f"Total rows    : {len(df):,}")
print(f"Countries     : {df['Country Code'].nunique()}")
print(f"Year range    : {df['Year'].min()} – {df['Year'].max()}")

print("\nFinal columns:")
for col in df.columns:
    print(f"  {col}")

print("\nMissing data (% missing per variable):")
missing = (df.isnull().mean() * 100).round(1)
for col, pct in missing.items():
    flag = " high" if pct > 30 else ""
    print(f"  {col:<30} {pct}%{flag}")

print("\nSample rows — USA:")
print(df[df['Country Code'] == 'USA'].tail(5).to_string(index=False))

# Save final dataset
df.to_csv('master_panel.csv', index=False)
print("\n master_panel.csv saved successfully.")
```

Appendix C: Missing Data Summary Output

The following table was generated directly from master_panel.csv using Python. The code used to produce this output is included in Appendix A.

Variable	Years Available	Missing (%)	Countries (any data)	Countries (80%+ coverage)	Countries (complete)
life_expectancy	2000-2023	0.43	232	232	232
edu_expenditure_pct_gdp	2000-2023	35.12	215	110	20
health_exp_pct_gdp	2000-2023	12.37	208	202	198
pm25	2001-2020	23.1	215	215	215
gdp_per_capita	2000-2022	32.55	164	164	164
le_women	2000-2023	7.73	215	215	215
le_men	2000-2023	7.73	215	215	215

Appendix D: Master Dataset

The cleaned panel dataset (master_panel.csv) covering 233 countries from 2000 to 2023 is submitted alongside this document. Each row represents one country-year observation containing all variables described in Section 3.3.

Appendix E: Generating visualizations

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.ticker as mticker
import seaborn as sns
from scipy import stats

df = pd.read_csv('master_panel.csv')

region_map = {
  'AFG':'South Asia','BGD':'South Asia','BTN':'South Asia','IND':'South Asia',
  'LKA':'South Asia','MDV':'South Asia','NPL':'South Asia','PAK':'South Asia',
  'AGO':'Sub-Saharan Africa','BEN':'Sub-Saharan Africa','BFA':'Sub-Saharan Africa',
  'BDI':'Sub-Saharan Africa','CMR':'Sub-Saharan Africa','CAF':'Sub-Saharan Africa',
```

```
'TCD':'Sub-Saharan Africa','COM':'Sub-Saharan Africa','COD':'Sub-Saharan Africa',
'COG':'Sub-Saharan Africa','CIV':'Sub-Saharan Africa','ETH':'Sub-Saharan Africa',
'GAB':'Sub-Saharan Africa','GMB':'Sub-Saharan Africa','GHA':'Sub-Saharan Africa',
'GIN':'Sub-Saharan Africa','GNB':'Sub-Saharan Africa','KEN':'Sub-Saharan Africa',
'LSO':'Sub-Saharan Africa','LBR':'Sub-Saharan Africa','MDG':'Sub-Saharan Africa',
'MWI':'Sub-Saharan Africa','MLI':'Sub-Saharan Africa','MRT':'Sub-Saharan Africa',
'MOZ':'Sub-Saharan Africa','NAM':'Sub-Saharan Africa','NER':'Sub-Saharan Africa',
'NGA':'Sub-Saharan Africa','RWA':'Sub-Saharan Africa','STP':'Sub-Saharan Africa',
'SEN':'Sub-Saharan Africa','SLE':'Sub-Saharan Africa','SOM':'Sub-Saharan Africa',
'ZAF':'Sub-Saharan Africa','SSD':'Sub-Saharan Africa','SDN':'Sub-Saharan Africa',
'SWZ':'Sub-Saharan Africa','TZA':'Sub-Saharan Africa','TGO':'Sub-Saharan Africa',
'UGA':'Sub-Saharan Africa','ZMB':'Sub-Saharan Africa','ZWE':'Sub-Saharan Africa',
'MUS':'Sub-Saharan Africa','CPV':'Sub-Saharan Africa','BWA':'Sub-Saharan Africa',
'ALB':'Europe & Central Asia','ARM':'Europe & Central Asia','AZE':'Europe & Central Asia',
'BLR':'Europe & Central Asia','BIH':'Europe & Central Asia','GEO':'Europe & Central Asia',
'KAZ':'Europe & Central Asia','KGZ':'Europe & Central Asia','MDA':'Europe & Central Asia',
'MNE':'Europe & Central Asia','MKD':'Europe & Central Asia','RUS':'Europe & Central Asia',
'SRB':'Europe & Central Asia','TJK':'Europe & Central Asia','TKM':'Europe & Central Asia',
'TUR':'Europe & Central Asia','UKR':'Europe & Central Asia','UZB':'Europe & Central Asia',
'AUT':'Europe & Central Asia','BEL':'Europe & Central Asia','BGR':'Europe & Central Asia',
'HRV':'Europe & Central Asia','CYP':'Europe & Central Asia','CZE':'Europe & Central Asia',
'DNK':'Europe & Central Asia','EST':'Europe & Central Asia','FIN':'Europe & Central Asia',
'FRA':'Europe & Central Asia','DEU':'Europe & Central Asia','GRC':'Europe & Central Asia',
'HUN':'Europe & Central Asia','ISL':'Europe & Central Asia','IRL':'Europe & Central Asia',
'ITA':'Europe & Central Asia','LVA':'Europe & Central Asia','LTU':'Europe & Central Asia',
'LUX':'Europe & Central Asia','NLD':'Europe & Central Asia','NOR':'Europe & Central Asia',
'POL':'Europe & Central Asia','PRT':'Europe & Central Asia','ROU':'Europe & Central Asia',
'SVK':'Europe & Central Asia','SVN':'Europe & Central Asia','ESP':'Europe & Central Asia',
'SWE':'Europe & Central Asia','CHE':'Europe & Central Asia','GBR':'Europe & Central Asia',
'DZA':'Middle East & North Africa','BHR':'Middle East & North Africa',
'EGY':'Middle East & North Africa','IRN':'Middle East & North Africa',
'IRQ':'Middle East & North Africa','ISR':'Middle East & North Africa',
'JOR':'Middle East & North Africa','KWT':'Middle East & North Africa',
'LBN':'Middle East & North Africa','LBY':'Middle East & North Africa',
'MAR':'Middle East & North Africa','OMN':'Middle East & North Africa',
'PSE':'Middle East & North Africa','QAT':'Middle East & North Africa',
'SAU':'Middle East & North Africa','SYR':'Middle East & North Africa',
'TUN':'Middle East & North Africa','ARE':'Middle East & North Africa',
'YEM':'Middle East & North Africa',
'ARG':'Latin America & Caribbean','BOL':'Latin America & Caribbean',
'BRA':'Latin America & Caribbean','CHL':'Latin America & Caribbean',
'COL':'Latin America & Caribbean','CRI':'Latin America & Caribbean',
'CUB':'Latin America & Caribbean','DOM':'Latin America & Caribbean',
'ECU':'Latin America & Caribbean','SLV':'Latin America & Caribbean',
```

```
    'GTM':'Latin America & Caribbean','HTI':'Latin America & Caribbean',
    'HND':'Latin America & Caribbean','JAM':'Latin America & Caribbean',
    'MEX':'Latin America & Caribbean','NIC':'Latin America & Caribbean',
    'PAN':'Latin America & Caribbean','PRY':'Latin America & Caribbean',
    'PER':'Latin America & Caribbean','TTO':'Latin America & Caribbean',
    'URY':'Latin America & Caribbean','VEN':'Latin America & Caribbean',
    'BLZ':'Latin America & Caribbean','GUY':'Latin America & Caribbean',
    'SUR':'Latin America & Caribbean',
    'CHN':'East Asia & Pacific','FJI':'East Asia & Pacific',
    'IDN':'East Asia & Pacific','KOR':'East Asia & Pacific',
    'LAO':'East Asia & Pacific','MYS':'East Asia & Pacific',
    'MNG':'East Asia & Pacific','MMR':'East Asia & Pacific',
    'NZL':'East Asia & Pacific','PNG':'East Asia & Pacific',
    'PHL':'East Asia & Pacific','SGP':'East Asia & Pacific',
    'TLS':'East Asia & Pacific','VNM':'East Asia & Pacific',
    'AUS':'East Asia & Pacific','KHM':'East Asia & Pacific',
    'JPN':'East Asia & Pacific','THA':'East Asia & Pacific',
    'USA':'North America','CAN':'North America',
}
df['region'] = df['Country Code'].map(region_map).fillna('Other')
plt.rcParams.update({
    'font.family': 'DejaVu Sans',
    'font.size': 11,
    'axes.titlesize': 13,
    'axes.titleweight': 'bold',
    'axes.spines.top': False,
    'axes.spines.right': False,
    'figure.dpi': 150
})

COLORS = {
    'main': '#2C6E9E',
    'accent': '#E05C2A',
    'women': '#C0392B',
    'men': '#2980B9',
    'grid': '#E8E8E8'
}

REGION_COLORS = {
    'East Asia & Pacific': '#E74C3C',
    'Europe & Central Asia': '#3498DB',
    'Latin America & Caribbean': '#2ECC71',
    'Middle East & North Africa': '#F39C12',
    'North America': '#9B59B6',
```

```
    'South Asia': '#E67E22',
    'Sub-Saharan Africa': '#1ABC9C',
    'Other': '#95A5A6'
}

# Generating here Figure 1: Global Life Expectancy Trend
fig, ax = plt.subplots(figsize=(10, 5))
global_trend = df.groupby('Year')['life_expectancy'].mean().reset_index()
ax.plot(global_trend['Year'], global_trend['life_expectancy'],
        color=COLORS['main'], linewidth=2.5, zorder=3)
ax.fill_between(global_trend['Year'], global_trend['life_expectancy'],
                alpha=0.12, color=COLORS['main'])
ax.axvspan(2020, 2021.5, alpha=0.12, color=COLORS['accent'], label='COVID-19 period')
ax.axvline(2020, color=COLORS['accent'], linestyle='--', linewidth=1.2, alpha=0.7)
ax.set_xlabel('Year')
ax.set_ylabel('Life Expectancy at Birth (years)')
ax.set_title('Figure 1. Global Average Life Expectancy Trend, 2000–2023')
ax.legend(frameon=False)
ax.yaxis.set_minor_locator(mticker.MultipleLocator(1))
ax.grid(axis='y', color=COLORS['grid'], zorder=0)
ax.set_xlim(2000, 2023)
plt.tight_layout()
# plt.savefig('/mnt/user-data/outputs/fig1_global_trend.png', bbox_inches='tight')
plt.savefig('fig1_global_trend.png', bbox_inches='tight')
plt.close()
print("Figure 1 saved")

# Generating here Figure 2: Life Expectancy by Region
fig, ax = plt.subplots(figsize=(11, 6))
regions_to_plot = [r for r in REGION_COLORS if r != 'Other']
for region in regions_to_plot:
#    print(df.columns.tolist())
    subset = df[df['region'] == region].groupby('Year')['life_expectancy'].mean()
    ax.plot(subset.index, subset.values,
            label=region, color=REGION_COLORS[region], linewidth=2)
ax.axvline(2020, color='gray', linestyle='--', linewidth=1.2, alpha=0.6, label='2020 (COVID-
19)')
ax.set_xlabel('Year')
ax.set_ylabel('Life Expectancy at Birth (years)')
ax.set_title('Figure 2. Life Expectancy Trends by Geographic Region, 2000–2023')
ax.legend(frameon=False, fontsize=9, loc='lower right')
ax.grid(axis='y', color=COLORS['grid'], zorder=0)
ax.set_xlim(2000, 2023)
plt.tight_layout()
```

```
# plt.savefig('/mnt/user-data/outputs/fig2_regional_trends.png', bbox_inches='tight')
plt.savefig('fig2_regional_trends.png', bbox_inches='tight')
plt.close()
print("Figure 2 saved")

# Generating here Figure 3: Preston Curve
fig, ax = plt.subplots(figsize=(10, 6))
scatter_df = df[df['log_gdp'].notna() & df['life_expectancy'].notna()].copy()
regions_present = scatter_df['region'].unique()
for region in regions_present:
  sub = scatter_df[scatter_df['region'] == region]
  color = REGION_COLORS.get(region, '#95A5A6')
  ax.scatter(sub['log_gdp'], sub['life_expectancy'],
        alpha=0.15, s=8, color=color, label=region)
# Fitted line
slope, intercept, r, p, se = stats.linregress(scatter_df['log_gdp'],
scatter_df['life_expectancy'])
x_range = np.linspace(scatter_df['log_gdp'].min(), scatter_df['log_gdp'].max(), 200)
ax.plot(x_range, intercept + slope * x_range,
    color='black', linewidth=2, zorder=5, label=f'Fitted line (R²={r**2:.2f})')
ax.set_xlabel('Log GDP per Capita')
ax.set_ylabel('Life Expectancy at Birth (years)')
ax.set_title('Figure 3. Life Expectancy vs. Log GDP per Capita (Preston Curve), 2000–2022')
ax.legend(frameon=False, fontsize=8, markerscale=2)
ax.grid(color=COLORS['grid'], zorder=0)
plt.tight_layout()
# plt.savefig('/mnt/user-data/outputs/fig3_preston_curve.png', bbox_inches='tight')
plt.savefig('fig3_preston_curve.png', bbox_inches='tight')
plt.close()
print("Figure 3 saved")

# Generating here Figure 4: Correlation Heatmap
fig, ax = plt.subplots(figsize=(9, 7))
corr_vars = ['life_expectancy', 'log_gdp', 'edu_expenditure_pct_gdp',
       'health_exp_pct_gdp', 'pm25']
corr_labels = ['Life Expectancy', 'Log GDP p.c.', 'Education Exp.', 'Health Exp.', 'PM2.5']
corr_df = df[corr_vars].dropna()
print(f"Correlation computed on {len(corr_df)} complete-case observations")
corr_matrix = corr_df.corr()
corr_matrix.index = corr_labels
corr_matrix.columns = corr_labels
mask = np.triu(np.ones_like(corr_matrix, dtype=bool), k=1)
sns.heatmap(corr_matrix, annot=True, fmt='.2f', cmap='RdBu_r',
      center=0, vmin=-1, vmax=1, ax=ax,
```

```
        linewidths=0.5, annot_kws={'size': 11})
ax.set_title('Figure 4. Correlation Heatmap of Key Variables')
plt.tight_layout()
# plt.savefig('/mnt/user-data/outputs/fig4_correlation_heatmap.png', bbox_inches='tight')
plt.savefig('fig4_correlation_heatmap.png', bbox_inches='tight')
plt.close()
print("Figure 4 saved")

# Generating here Figure 5: Women vs Men Life Expectancy
fig, ax = plt.subplots(figsize=(10, 5))
# women_trend = df.groupby('Year')['le_women'].mean()
# men_trend = df.groupby('Year')['le_men'].mean()
sex_df = df.groupby('Year')[['le_women','le_men']].mean().dropna()
women_trend = sex_df['le_women']
men_trend = sex_df['le_men']
ax.plot(women_trend.index, women_trend.values,
     color=COLORS['women'], linewidth=2.5, label='Women')
ax.plot(men_trend.index, men_trend.values,
     color=COLORS['men'], linewidth=2.5, label='Men')
ax.fill_between(women_trend.index, men_trend.values, women_trend.values,
          alpha=0.1, color='purple', label='Gender gap')
ax.axvline(2020, color='gray', linestyle='--', linewidth=1.2, alpha=0.6)
ax.set_xlabel('Year')
ax.set_ylabel('Life Expectancy at Birth (years)')
ax.set_title('Figure 5. Life Expectancy Trends by Sex, 2000–2023')
ax.legend(frameon=False)
ax.grid(axis='y', color=COLORS['grid'], zorder=0)
ax.set_xlim(2000, 2023)
plt.tight_layout()
# plt.savefig('/mnt/user-data/outputs/fig5_sex_comparison.png', bbox_inches='tight')
plt.savefig('fig5_sex_comparison.png', bbox_inches='tight')
plt.close()
print("Figure 5 saved")

# Generating here Figure 6: Pre vs Post 2020 COVID Comparison
fig, axes = plt.subplots(1, 2, figsize=(12, 5))
pre = df[df['Year'] < 2020]['life_expectancy'].dropna()
post = df[df['Year'] >= 2020]['life_expectancy'].dropna()
axes[0].hist(pre, bins=30, color=COLORS['main'], alpha=0.7, edgecolor='white')
axes[0].axvline(pre.mean(), color=COLORS['accent'], linewidth=2,
          label=f'Mean: {pre.mean():.1f}')
axes[0].set_title('Pre-2020 (2000–2019)')
axes[0].set_xlabel('Life Expectancy (years)')
axes[0].set_ylabel('Frequency')
```

```
axes[0].legend(frameon=False)
axes[0].grid(axis='y', color=COLORS['grid'])

axes[1].hist(post, bins=30, color=COLORS['accent'], alpha=0.7, edgecolor='white')
axes[1].axvline(post.mean(), color=COLORS['main'], linewidth=2,
        label=f'Mean: {post.mean():.1f}')
axes[1].set_title('2020–2023 (COVID-19 Period)')
axes[1].set_xlabel('Life Expectancy (years)')
axes[1].set_ylabel('Frequency')
axes[1].legend(frameon=False)
axes[1].grid(axis='y', color=COLORS['grid'])

fig.suptitle('Figure 6. Distribution of Life Expectancy: Pre-2020 vs. 2020–2023',
       fontweight='bold', fontsize=13)
plt.tight_layout()
# plt.savefig('/mnt/user-data/outputs/fig6_covid_comparison.png', bbox_inches='tight')
plt.savefig('fig6_covid_comparison.png', bbox_inches='tight')
plt.close()
print("Figure 6 saved")

print("\n All 6 figures generated successfully!")
```

Appendix F: Verifying stats

```
import pandas as pd
import numpy as np
from scipy import stats

df = pd.read_csv('master_panel.csv')

region_map = {
  'AFG':'South Asia','BGD':'South Asia','BTN':'South Asia','IND':'South Asia',
  'LKA':'South Asia','MDV':'South Asia','NPL':'South Asia','PAK':'South Asia',
  'AGO':'Sub-Saharan Africa','BEN':'Sub-Saharan Africa','BFA':'Sub-Saharan Africa',
  'BDI':'Sub-Saharan Africa','CMR':'Sub-Saharan Africa','CAF':'Sub-Saharan Africa',
  'TCD':'Sub-Saharan Africa','COM':'Sub-Saharan Africa','COD':'Sub-Saharan Africa',
  'COG':'Sub-Saharan Africa','CIV':'Sub-Saharan Africa','ETH':'Sub-Saharan Africa',
  'GAB':'Sub-Saharan Africa','GMB':'Sub-Saharan Africa','GHA':'Sub-Saharan Africa',
  'GIN':'Sub-Saharan Africa','GNB':'Sub-Saharan Africa','KEN':'Sub-Saharan Africa',
  'LSO':'Sub-Saharan Africa','LBR':'Sub-Saharan Africa','MDG':'Sub-Saharan Africa',
  'MWI':'Sub-Saharan Africa','MLI':'Sub-Saharan Africa','MRT':'Sub-Saharan Africa',
  'MOZ':'Sub-Saharan Africa','NAM':'Sub-Saharan Africa','NER':'Sub-Saharan Africa',
  'NGA':'Sub-Saharan Africa','RWA':'Sub-Saharan Africa','STP':'Sub-Saharan Africa',
  'SEN':'Sub-Saharan Africa','SLE':'Sub-Saharan Africa','SOM':'Sub-Saharan Africa',
```

```
'ZAF':'Sub-Saharan Africa','SSD':'Sub-Saharan Africa','SDN':'Sub-Saharan Africa',
'SWZ':'Sub-Saharan Africa','TZA':'Sub-Saharan Africa','TGO':'Sub-Saharan Africa',
'UGA':'Sub-Saharan Africa','ZMB':'Sub-Saharan Africa','ZWE':'Sub-Saharan Africa',
'MUS':'Sub-Saharan Africa','CPV':'Sub-Saharan Africa','BWA':'Sub-Saharan Africa',
'ALB':'Europe & Central Asia','ARM':'Europe & Central Asia','AZE':'Europe & Central Asia',
'BLR':'Europe & Central Asia','BIH':'Europe & Central Asia','GEO':'Europe & Central Asia',
'KAZ':'Europe & Central Asia','KGZ':'Europe & Central Asia','MDA':'Europe & Central Asia',
'MNE':'Europe & Central Asia','MKD':'Europe & Central Asia','RUS':'Europe & Central Asia',
'SRB':'Europe & Central Asia','TJK':'Europe & Central Asia','TKM':'Europe & Central Asia',
'TUR':'Europe & Central Asia','UKR':'Europe & Central Asia','UZB':'Europe & Central Asia',
'AUT':'Europe & Central Asia','BEL':'Europe & Central Asia','BGR':'Europe & Central Asia',
'HRV':'Europe & Central Asia','CYP':'Europe & Central Asia','CZE':'Europe & Central Asia',
'DNK':'Europe & Central Asia','EST':'Europe & Central Asia','FIN':'Europe & Central Asia',
'FRA':'Europe & Central Asia','DEU':'Europe & Central Asia','GRC':'Europe & Central Asia',
'HUN':'Europe & Central Asia','ISL':'Europe & Central Asia','IRL':'Europe & Central Asia',
'ITA':'Europe & Central Asia','LVA':'Europe & Central Asia','LTU':'Europe & Central Asia',
'LUX':'Europe & Central Asia','NLD':'Europe & Central Asia','NOR':'Europe & Central Asia',
'POL':'Europe & Central Asia','PRT':'Europe & Central Asia','ROU':'Europe & Central Asia',
'SVK':'Europe & Central Asia','SVN':'Europe & Central Asia','ESP':'Europe & Central Asia',
'SWE':'Europe & Central Asia','CHE':'Europe & Central Asia','GBR':'Europe & Central Asia',
'DZA':'Middle East & North Africa','BHR':'Middle East & North Africa',
'EGY':'Middle East & North Africa','IRN':'Middle East & North Africa',
'IRQ':'Middle East & North Africa','ISR':'Middle East & North Africa',
'JOR':'Middle East & North Africa','KWT':'Middle East & North Africa',
'LBN':'Middle East & North Africa','LBY':'Middle East & North Africa',
'MAR':'Middle East & North Africa','OMN':'Middle East & North Africa',
'PSE':'Middle East & North Africa','QAT':'Middle East & North Africa',
'SAU':'Middle East & North Africa','SYR':'Middle East & North Africa',
'TUN':'Middle East & North Africa','ARE':'Middle East & North Africa',
'YEM':'Middle East & North Africa',
'ARG':'Latin America & Caribbean','BOL':'Latin America & Caribbean',
'BRA':'Latin America & Caribbean','CHL':'Latin America & Caribbean',
'COL':'Latin America & Caribbean','CRI':'Latin America & Caribbean',
'CUB':'Latin America & Caribbean','DOM':'Latin America & Caribbean',
'ECU':'Latin America & Caribbean','SLV':'Latin America & Caribbean',
'GTM':'Latin America & Caribbean','HTI':'Latin America & Caribbean',
'HND':'Latin America & Caribbean','MEX':'Latin America & Caribbean',
'NIC':'Latin America & Caribbean','PAN':'Latin America & Caribbean',
'PRY':'Latin America & Caribbean','PER':'Latin America & Caribbean',
'TTO':'Latin America & Caribbean','URY':'Latin America & Caribbean',
'VEN':'Latin America & Caribbean','BLZ':'Latin America & Caribbean',
'GUY':'Latin America & Caribbean','SUR':'Latin America & Caribbean',
'CHN':'East Asia & Pacific','FJI':'East Asia & Pacific',
'IDN':'East Asia & Pacific','KOR':'East Asia & Pacific',
```

```
    'LAO':'East Asia & Pacific','MYS':'East Asia & Pacific',
    'MNG':'East Asia & Pacific','MMR':'East Asia & Pacific',
    'NZL':'East Asia & Pacific','PNG':'East Asia & Pacific',
    'PHL':'East Asia & Pacific','SGP':'East Asia & Pacific',
    'TLS':'East Asia & Pacific','VNM':'East Asia & Pacific',
    'AUS':'East Asia & Pacific','KHM':'East Asia & Pacific',
    'JPN':'East Asia & Pacific','THA':'East Asia & Pacific',
    'USA':'North America','CAN':'North America',
}
df['region'] = df['Country Code'].map(region_map).fillna('Other')

print("Figure 1 Stats")
global_trend = df.groupby('Year')['life_expectancy'].mean()
for yr in [2000, 2019, 2020, 2021, 2022, 2023]:
    print(f"{yr}: {global_trend[yr]:.2f}")

print("\nFigure 2 Regional 2022")
for region in sorted(df['region'].unique()):
    val = df[(df['region']==region)&(df['Year']==2022)]['life_expectancy'].mean()
    n = df[(df['region']==region)&(df['Year']==2022)]['Country Code'].nunique()
    print(f"{region}: {val:.1f} ({n} countries)")

print("\nFigure 3 Preston Curve")
scatter_df = df[df['log_gdp'].notna() & df['life_expectancy'].notna()]
slope, intercept, r, p, se = stats.linregress(scatter_df['log_gdp'],
scatter_df['life_expectancy'])
print(f"R^2: {r**2:.3f}, Slope: {slope:.3f}, N: {len(scatter_df)}")

print("\nFigure 4 Correlations (complete cases)")
corr_vars =
['life_expectancy','log_gdp','edu_expenditure_pct_gdp','health_exp_pct_gdp','pm25']
corr_df = df[corr_vars].dropna()
print(f"Complete cases used: {len(corr_df)}")
print(corr_df.corr()['life_expectancy'].round(3))

print("\nFigure 5 Sex Stats")
women = df.groupby('Year')['le_women'].mean()
men = df.groupby('Year')['le_men'].mean()
for yr in [2000, 2023]:
    print(f"{yr} — Women: {women[yr]:.1f}, Men: {men[yr]:.1f}, Gap: {women[yr]-men[yr]:.1f}")

print("\nFigure 6 COVID Stats")
pre = df[df['Year']<2020]['life_expectancy'].mean()
post = df[df['Year']>=2020]['life_expectancy'].mean()
```

```
print(f"Pre-2020 mean: {pre:.2f}")
print(f"Post-2020 mean: {post:.2f}")
```

Appendix G: Regression results

```
from statsmodels.stats.outliers_influence import variance_inflation_factor
import pandas as pd
import numpy as np
import statsmodels.formula.api as smf
import warnings
warnings.filterwarnings('ignore')

#  Load the Data
df = pd.read_csv('master_panel.csv')

print("=" * 65)
print("REGRESSION ANALYSIS: DETERMINANTS OF GLOBAL LIFE EXPECTANCY")
print("=" * 65)

# MODEL 1: Baseline is LE vs log GDP plus Year
# Only countries with log_gdp data
m1_data = df[['Country Code', 'life_expectancy', 'log_gdp', 'Year']].dropna()

model1 = smf.ols(
   'life_expectancy ~ log_gdp + Year',
   data=m1_data
).fit(cov_type='HC1')  # robust standard errors

print("\n" + "=" * 65)
print("MODEL 1: Baseline (Life Expectancy ~ Log GDP + Year)")
print(
   f"Observations: {int(model1.nobs):,} | Countries: {m1_data['Country Code'].nunique()}")
print("=" * 65)
print(f"{'Variable':<30} {'Coef':>8} {'Std Err':>10} {'p-value':>10}")
print("-" * 65)
for var in ['Intercept', 'log_gdp', 'Year']:
   coef = model1.params[var]
   se = model1.bse[var]
   pval = model1.pvalues[var]
   stars = '***' if pval < 0.01 else '**' if pval < 0.05 else '*' if pval < 0.1 else ''
   print(f"{var:<30} {coef:>8.3f} {se:>10.3f} {pval:>10.4f} {stars}")
print(f"\nR²: {model1.rsquared:.3f} | Adj. R²: {model1.rsquared_adj:.3f}")
```

```
# MODEL 2: Add Education plus Health Expenditure

m2_data = df[['Country Code', 'life_expectancy', 'log_gdp', 'Year',
          'edu_expenditure_pct_gdp', 'health_exp_pct_gdp']].dropna()

model2 = smf.ols(
   'life_expectancy ~ log_gdp + Year + edu_expenditure_pct_gdp + health_exp_pct_gdp',
   data=m2_data
).fit(cov_type='HC1')

print("\n" + "=" * 65)
print("MODEL 2: + Education + Health Expenditure")
print(
   f"Observations: {int(model2.nobs):,} | Countries: {m2_data['Country Code'].nunique()}")
print("=" * 65)
print(f"{'Variable':<30} {'Coef':>8} {'Std Err':>10} {'p-value':>10}")
print("-" * 65)
for var in ['Intercept', 'log_gdp', 'Year', 'edu_expenditure_pct_gdp', 'health_exp_pct_gdp']:
   coef = model2.params[var]
   se = model2.bse[var]
   pval = model2.pvalues[var]
   stars = '***' if pval < 0.01 else '**' if pval < 0.05 else '*' if pval < 0.1 else ''
   print(f"{var:<30} {coef:>8.3f} {se:>10.3f} {pval:>10.4f} {stars}")
print(f"\nR²: {model2.rsquared:.3f} | Adj. R²: {model2.rsquared_adj:.3f}")

# MODEL 3: Add PM2.5 (2001-2020 only)

m3_data = df[
   (df['Year'] >= 2001) & (df['Year'] <= 2020)
][['Country Code', 'life_expectancy', 'log_gdp', 'Year',
  'edu_expenditure_pct_gdp', 'health_exp_pct_gdp', 'pm25']].dropna()

model3 = smf.ols(
   'life_expectancy ~ log_gdp + Year + edu_expenditure_pct_gdp + health_exp_pct_gdp +
pm25',
   data=m3_data
).fit(cov_type='HC1')

print("\n" + "=" * 65)
print("MODEL 3: + PM2.5 (subsample 2001-2020)")

print(
   f"Observations: {int(model3.nobs):,} | Countries: {m3_data['Country Code'].nunique()}")
```

```
print("=" * 65)
print(f"{'Variable':<30} {'Coef':>8} {'Std Err':>10} {'p-value':>10}")
print("-" * 65)
for var in ['Intercept', 'log_gdp', 'Year', 'edu_expenditure_pct_gdp',
        'health_exp_pct_gdp', 'pm25']:
    coef = model3.params[var]
    se = model3.bse[var]
    pval = model3.pvalues[var]
    stars = '***' if pval < 0.01 else '**' if pval < 0.05 else '*' if pval < 0.1 else ''
    print(f"{var:<30} {coef:>8.3f} {se:>10.3f} {pval:>10.4f} {stars}")
print(f"\nR²: {model3.rsquared:.3f} | Adj. R²: {model3.rsquared_adj:.3f}")

#  VIF Check

print("\n" + "=" * 65)
print("MULTICOLLINEARITY CHECK (VIF) - Model 3 Predictors")
print("=" * 65)
vif_data = m3_data[['log_gdp', 'Year', 'edu_expenditure_pct_gdp',
             'health_exp_pct_gdp', 'pm25']].dropna()
vif_data = (vif_data - vif_data.mean()) / vif_data.std()
vif_results = pd.DataFrame({
    'Variable': vif_data.columns,
    'VIF': [variance_inflation_factor(vif_data.values, i)
        for i in range(vif_data.shape[1])]
})
print(vif_results.round(2).to_string(index=False))
print("\nNote: VIF > 10 indicates serious multicollinearity")

# Save results
results = {
    'Model': ['Model 1', 'Model 2', 'Model 3'],
    'N_obs': [int(model1.nobs), int(model2.nobs), int(model3.nobs)],
    'R2': [round(model1.rsquared, 3), round(model2.rsquared, 3), round(model3.rsquared,
3)],
    'Adj_R2': [round(model1.rsquared_adj, 3), round(model2.rsquared_adj, 3),
round(model3.rsquared_adj, 3)]
}
pd.DataFrame(results).to_csv('regression_summary.csv', index=False)
print("\n Regression analysis complete. Results saved.")
```

Appendix H: Robustness Checks

```
import pandas as pd
import numpy as np
import statsmodels.api as sm
from statsmodels.stats.outliers_influence import variance_inflation_factor

# Robustness Checks
# Inputs: master_panel.csv (created by your merge/clean script)
# Outputs:
#   - robustness_results.csv
#   - vif_model3.csv

DATA_FILE = "master_panel.csv"

# Helper: running OLS with HC1 SE

def run_ols(df_in, y_col, x_cols):
  X = df_in[x_cols].copy()
  X = sm.add_constant(X)
  y = df_in[y_col]
  model = sm.OLS(y, X).fit(cov_type="HC1")
  return model

def safe_get(model, name):
  """Return (coef, pvalue) if exists else (None, None)."""
  if name in model.params.index:
    return float(model.params[name]), float(model.pvalues[name])
  return None, None

# Loads data
df = pd.read_csv(DATA_FILE)

# Basic sanity filters (optional but safe)
df = df[df["Year"].between(2000, 2023)]
df = df[df["Country Code"].astype(str).str.len() == 3]  # ISO3 only

# CHECK 1: Pre-2020 re-runs (Model 1 and Model 2)

df_pre = df[df["Year"] <= 2019].copy()

# Model 1 pre-2020
m1_pre = df_pre[["Country Code",
          "life_expectancy", "log_gdp", "Year"]].dropna()
```

```
r1 = run_ols(m1_pre, "life_expectancy", ["log_gdp", "Year"])

# Model 2 pre-2020
m2_pre = df_pre[[
  "Country Code", "life_expectancy", "log_gdp", "Year",
  "edu_expenditure_pct_gdp", "health_exp_pct_gdp"
]].dropna()
r2 = run_ols(m2_pre, "life_expectancy", [
       "log_gdp", "Year", "edu_expenditure_pct_gdp", "health_exp_pct_gdp"])

# CHECK 2: GDP outlier trimming (Model 1 trimmed 1% tails)

g = df["gdp_per_capita"].dropna()
low, high = g.quantile(0.01), g.quantile(0.99)

df_trim = df[df["gdp_per_capita"].between(low, high)].copy()
m1_trim = df_trim[["Country Code",
          "life_expectancy", "log_gdp", "Year"]].dropna()
r1t = run_ols(m1_trim, "life_expectancy", ["log_gdp", "Year"])

# CHECK 3: VIF for Model 3 predictors (2001–2020 complete cases)

m3 = df[df["Year"].between(2001, 2020)][[
  "life_expectancy", "log_gdp", "Year",
  "edu_expenditure_pct_gdp", "health_exp_pct_gdp", "pm25"
]].dropna()

X_vif = m3[["log_gdp", "Year", "edu_expenditure_pct_gdp",
       "health_exp_pct_gdp", "pm25"]].copy()
X_vif = sm.add_constant(X_vif)

vif_rows = []
for i, col in enumerate(X_vif.columns):
  if col == "const":
    continue
  vif_rows.append({
    "Variable": col,
    "VIF": round(variance_inflation_factor(X_vif.values, i), 2)
  })

vif_df = pd.DataFrame(vif_rows)
vif_df.to_csv("vif_model3.csv", index=False)

# Saving robustness summary
```

```
robust_rows = []

# Pre-2020 Model 1
coef, p = safe_get(r1, "log_gdp")
robust_rows.append({
    "Check": "Pre2020_Model1",
    "N_obs": int(r1.nobs),
    "Countries": int(m1_pre["Country Code"].nunique()),
    "R2": round(r1.rsquared, 3),
    "Adj_R2": round(r1.rsquared_adj, 3),
    "log_gdp_coef": round(coef, 3) if coef is not None else None,
    "log_gdp_p": round(p, 4) if p is not None else None,
    "health_exp_coef": None,
    "health_exp_p": None,
    "edu_exp_coef": None,
    "edu_exp_p": None
})

# Pre-2020 Model 2
coef_g, p_g = safe_get(r2, "log_gdp")
coef_h, p_h = safe_get(r2, "health_exp_pct_gdp")
coef_e, p_e = safe_get(r2, "edu_expenditure_pct_gdp")

robust_rows.append({
    "Check": "Pre2020_Model2",
    "N_obs": int(r2.nobs),
    "Countries": int(m2_pre["Country Code"].nunique()),
    "R2": round(r2.rsquared, 3),
    "Adj_R2": round(r2.rsquared_adj, 3),
    "log_gdp_coef": round(coef_g, 3) if coef_g is not None else None,
    "log_gdp_p": round(p_g, 4) if p_g is not None else None,
    "health_exp_coef": round(coef_h, 3) if coef_h is not None else None,
    "health_exp_p": round(p_h, 4) if p_h is not None else None,
    "edu_exp_coef": round(coef_e, 3) if coef_e is not None else None,
    "edu_exp_p": round(p_e, 4) if p_e is not None else None
})

# Trimmed GDP Model 1
coef_t, p_t = safe_get(r1t, "log_gdp")
robust_rows.append({
    "Check": "TrimmedGDP_Model1_1pct",
    "N_obs": int(r1t.nobs),
    "Countries": int(m1_trim["Country Code"].nunique()),
```

```
    "R2": round(r1t.rsquared, 3),
    "Adj_R2": round(r1t.rsquared_adj, 3),
    "log_gdp_coef": round(coef_t, 3) if coef_t is not None else None,
    "log_gdp_p": round(p_t, 4) if p_t is not None else None,
    "health_exp_coef": None,
    "health_exp_p": None,
    "edu_exp_coef": None,
    "edu_exp_p": None
})

robust_df = pd.DataFrame(robust_rows)
robust_df.to_csv("robustness_results.csv", index=False)

# Console output (for logs)

print("\n ROBUSTNESS SUMMARY SAVED")
print(robust_df.to_string(index=False))

print("\n VIF SAVED (Model 3 predictors) ")
print(vif_df.to_string(index=False))

print("\n Robustness checks complete.")
```